

Displays

D1:

$$V(x, t) = \sup_{u(\cdot)} J_{u(\cdot)}(x, t) \quad (1)$$

D2:

$$\sum_{i=0}^k \binom{N}{i} \epsilon^i (1 - \epsilon)^{N-i} \leq \beta \quad (2)$$

D3:

$$\mathbb{P}_{x \in \mathcal{S}} (V(x, 0) \leq 0) \leq \epsilon \quad (3)$$

D4:

$$\mathbb{P}_{x \in \mathcal{S}} (J_{\tilde{\pi}}(x, 0) > 0) \sim \text{Beta}(N - k, k + 1) \quad (4)$$

D5:

$$\sum_{i=0}^k \binom{N}{i} \epsilon^i (1 - \epsilon)^{N-i} \leq \beta \quad (5)$$

D6:

$$\mathbb{P}_{x \in \mathcal{S}} (V(x, 0) \leq 0) \leq \epsilon \quad (6)$$

D7:

$$\begin{aligned} \text{CCP}_{\epsilon} : \min_{g \in \mathbb{R}} g \\ \text{s.t. } \mathbb{P}_{h \in H} (f(h) \leq g) \geq 1 - \epsilon \end{aligned} \quad (7)$$

D8:

$$\begin{aligned} \text{SP}_{N,k}^A : \min_{g \in \mathbb{R}} g \\ \text{s.t. } f(h_i) \leq g, \quad i \in \{1, \dots, N\} - A\{h_1, \dots, h_N\} \end{aligned} \quad (8)$$

D9:

$$\sum_{i=0}^k \binom{N}{i} \epsilon^i (1 - \epsilon)^{N-i} \leq \beta \quad (9)$$

D10:

$$\mathbb{P}_{h \in H} (f(h) > g_{N,k}^*) \leq \epsilon \quad (10)$$

D11:

$$\mathbb{P}_{(x_{1:N}, x) \in \mathcal{S}} (-J_{\tilde{\pi}}(x, 0) \leq \hat{q}) \geq 1 - \alpha$$

D12:

$$\mathbb{P}_{(x_{1:N}, x) \in \mathcal{S}} (J_{\tilde{\pi}}(x, 0) \geq -\hat{q}) \geq 1 - \frac{k+1}{N+1}$$

D13:

$$\mathbb{P}_{(x_{1:N}, x) \in \mathcal{S}} (J_{\tilde{\pi}}(x, 0) > 0) \geq \frac{N-k}{N+1} \quad (11)$$

D14:

$$\mathbb{P}_{x \in \mathcal{S}} (J_{\tilde{\pi}}(x, 0) > 0) \sim \text{Beta}(N + 1 - l, l), \quad l = \lfloor (N + 1)\alpha \rfloor$$

D15:

$$\mathbb{P}_{x \in \mathcal{S}} (J_{\tilde{\pi}}(x, 0) > 0) \sim \text{Beta}(N - k, k + 1) \quad (12)$$

D16:

$$\mathbb{P}(Y_{\text{test}} \in C(X_{\text{test}}) | \{(X_i, Y_i)\}_{i=1}^n) \sim \text{Beta}(n + 1 - l, l), \quad l = \lfloor (n + 1)\alpha \rfloor \quad (13)$$

D17:

$$\mathbb{P}(Y_{\text{test}} \in C(X_{\text{test}}) | \{(X_i, Y_i)\}_{i=1}^n) \geq 1 - \epsilon \quad (14)$$

Inline

- I1: $x \in X \subseteq \mathbb{R}^n$
- I2: $u \in \mathcal{U}$
- I3: $\dot{x} = f(x, u)$
- I4: x
- I5: T
- I6: $\xi_{x,t}^u(\tau)$
- I7: $\tau \in [t, T]$
- I8: t
- I9: $u(\cdot)$
- I10: $[t, \tau]$
- I11: $\mathcal{L}$
- I12: Q_1, Q_2, Q_3
- I13: Q_i
- I14: (p_{xi}, p_{yi})
- I15: θ_i
- I16: v
- I17: $u_i \in [u_{\min}, u_{\max}]$
- I18: $\dot{p}_{xi} = v \cos \theta_i, \quad \dot{p}_{yi} = v \sin \theta_i, \quad \dot{\theta}_i = u_i$
- I19: $\mathcal{L} = \{x : \min\{d(Q_1, Q_2), d(Q_1, Q_3), d(Q_2, Q_3)\} \leq R\}$
- I20: $d(Q_i, Q_j)$
- I21: Q_j
- I22: $v = 0.6, \quad u_{\min} = -1.1, \quad u_{\max} = 1.1, \quad R = 0.25$
- I23: X
- I24: $[0, T]$
- I25: $\text{BRT} = \{x : x \in X, \forall u(\cdot), \exists \tau \in [0, T], \xi_{x,0}^u(\tau) \in \mathcal{L}\}$
- I26: $\mathcal{S}$
- I27: $\mathbb{P}_{x \in \mathcal{S}}(x \in \text{BRT}) \leq \epsilon$
- I28: $\epsilon \in (0, 1)$
- I29: $\mathbb{P}_{x \in \mathcal{S}}(x \in \text{BRT}^C) \leq \epsilon$
- I30: $l(x)$
- I31: $\mathcal{L} = \{x : l(x) \leq 0\}$
- I32: $l(x) = \min\{d(Q_1, Q_2), d(Q_1, Q_3), d(Q_2, Q_3)\} - R$
- I33: $J_{u(\cdot)}(x, t) = \min_{\tau \in [t, T]} l(\xi_{x,t}^u(\tau))$
- I34: $J_{u(\cdot)}(x, t)$
- I35: $\text{BRT} = \{x : x \in X, V(x, 0) \leq 0\}$
- I36: $\min \left\{ D_t V(x, t) + H(x, t), l(x) - V(x, t) \right\} = 0$
- I37: $V(x, T) = l(x)$
- I38: D_t
- I39: ∇
- I40: V
- I41: H
- I42: $H(x, t) = \max_u \langle \nabla V(x, t), f(x, u) \rangle$
- I43: $u^*(x, t) = \arg \max_u \langle \nabla V(x, t), f(x, u) \rangle$
- I44: $\tilde{V}(x, t)$
- I45: $\tilde{\pi}(x, t)$
- I46: $\delta_{\tilde{V}, \tilde{\pi}} := \max_{x \in X} \{\tilde{V}(x, 0) : J_{\tilde{\pi}}(x, 0) \leq 0\}$
- I47: $\delta_{\tilde{V}, \tilde{\pi}}$
- I48: $\tilde{V}(x, 0)$
- I49: $\mathcal{S} \subseteq X$
- I50: δ
- I51: $\delta > 0$
- I52: N
- I53: $x_{1:N}$

I54: $\mathbb{P}$
 I55: $J_{\tilde{\pi}}(x_i, 0)$
 I56: $i = 1, 2, \dots, N$
 I57: x_i
 I58: k
 I59: $J_{\tilde{\pi}}(x_i, 0) \leq 0$
 I60: $\beta \in (0, 1)$
 I61: $1 - \beta$
 I62: 1
 I63: β
 I64: ϵ
 I65: 0
 I66: 10^{-16}
 I67: 1
 I68: $\beta = 10^{-16}$
 I69: $-J_{\tilde{\pi}}(x, 0)$
 I70: $N - k$
 I71: $k + 1$
 I72: $N : k$
 I73: $\mathbb{P}_{x \in \mathcal{S}} (J_{\tilde{\pi}}(x, 0) > 0)$
 I74: $k = 731$
 I75: $N = 3684118$
 I76: $1 - \beta = 0.9$
 I77: $1 - \epsilon = 0.99979$
 I78: $\frac{N-k}{N+1}$
 I79: $\mathbb{P}_{(x_{1:N}, x) \in \mathcal{S}} (J_{\tilde{\pi}}(x, 0) > 0) \geq \frac{N-k}{N+1}$
 I80: $\tilde{V}(x_i, 0) \geq \delta$
 I81: $J_{\tilde{\pi}}(x_i, 0) \leq 0$
 I82: $J_{\tilde{\pi}}(x, 0)$
 I83: $\tilde{J}_{\tilde{\pi}}(x, 0)$
 I84: $\mathcal{T}$
 I85: n
 I86: $\mathcal{T} = (x_1, J_{\tilde{\pi}}(x_1, 0)), \dots, (x_n, J_{\tilde{\pi}}(x_n, 0))$
 I87: $\frac{1}{n} \sum_{i=1}^n w_i (\tilde{V}(x_i, 0) - J_{\tilde{\pi}}(x_i, 0))^2$
 I88: $w_i = w$
 I89: $\left(\tilde{V}(x_i, 0) < J_{\tilde{\pi}}(x_i, 0) \right)$
 I90: $w_i = 1$
 I91: w
 I92: $\left(\tilde{V}(x_i, 0) > J_{\tilde{\pi}}(x_i, 0) \right)$
 I93: $\mathcal{V}$
 I94: $\max_{x \in \mathcal{V}} \{ \tilde{J}_{\tilde{\pi}}(x, 0) : J_{\tilde{\pi}}(x, 0) \leq 0 \}$
 I95: $w = 10^{-3}$
 I96: $\epsilon \leq 10^{-4}$
 I97: $\epsilon = 10^{-4}$
 I98: (p_x, p_y)
 I99: θ
 I100: (v_x, v_y)
 I101: ω
 I102: $\tau_1, \tau_2 \in [-250, 250]$
 I103: $\dot{p}_x = v_x, \dot{p}_y = v_y, \dot{\theta} = \omega, \dot{\omega} = 0.3\tau_1, \dot{v}_x = \tau_1 \cos \theta - \tau_2 \sin \theta, \dot{v}_y = \tau_1 \sin \theta + \tau_2 \cos \theta - g$
 I104: $g = 9.81$
 I105: $\mathcal{L} = \{x : |p_x| < 20.0, p_y < 20.0\}$
 I106: g
 I107: h
 I108: f

I109: A
 I110: $g_{N,k}^*$
 I111: $\mathbb{R}$
 I112: $\{g : f(h) \leq g\}$
 I113: $d = 1$
 I114: $c = 1$
 I115: $x = g$
 I116: $X = G = \mathbb{R}$
 I117: $\delta = h$
 I118: $\Delta = H$
 I119: $X_\delta = G_h = \{g : f(h) \leq g\}$
 I120: $\blacksquare$
 I121: $h = x$
 I122: $H = \mathcal{S}$
 I123: $f(h) = f(x) = -J_{\bar{\pi}}(x, 0)$
 I124: $f(h_i) \leq g$
 I125: $f(h_i) = -J_{\bar{\pi}}(x_i, 0) \geq 0$
 I126: $f(h_i) < 0$
 I127: < 0
 I128: $0 < -g_{N,k}^*$
 I129: $\mathbb{P}_{x \in \mathcal{S}}(J_{\bar{\pi}}(x, 0) \leq 0) \leq \mathbb{P}_{x \in \mathcal{S}}(J_{\bar{\pi}}(x, 0) < -g_{N,k}^*)$
 I130: $\mathbb{P}_{x \in \mathcal{S}}(J_{\bar{\pi}}(x, 0) < -g_{N,k}^*) \leq \epsilon \implies \mathbb{P}_{x \in \mathcal{S}}(J_{\bar{\pi}}(x, 0) \leq 0) \leq \epsilon$
 I131: $\forall(x, t), J_{\bar{\pi}}(x, t) \leq V(x, t)$
 I132: $x = x$
 I133: $y = -J_{\bar{\pi}}(x, 0)$
 I134: $s(x, y) = y = -J_{\bar{\pi}}(x, 0)$
 I135: $n = N$
 I136: $\alpha = \frac{k+1}{N+1}$
 I137: $\hat{q}$
 I138: $\frac{\lceil (N+1)(1-\alpha) \rceil}{N}$
 I139: $-J_{\bar{\pi}}(x_{1:N}, 0)$
 I140: $\frac{\lceil (N+1)(1-\alpha) \rceil}{N} = \frac{\lceil (N+1)(1-\frac{k+1}{N+1}) \rceil}{N} = \frac{\lceil (N+1)(\frac{N-k}{N+1}) \rceil}{N} = \frac{N-k}{N}$
 I141: $-J_{\bar{\pi}}(x_i, 0) \geq 0$
 I142: $\hat{q} < 0$
 I143: $J_{\bar{\pi}}(x, 0) \geq -\hat{q}$
 I144: $J_{\bar{\pi}}(x, 0) > 0$
 I145: $s(x, y) \in \mathbb{R}$
 I146: y
 I147: $(X_1, Y_1), \dots, (X_n, Y_n)$
 I148: $\frac{\lceil (n+1)(1-\alpha) \rceil}{n}$
 I149: $s(X_1, Y_1), \dots, s(X_n, Y_n)$
 I150: $\alpha \in [0, 1]$
 I151: X_{test}
 I152: $C(X_{\text{test}}) = \{y : s(X_{\text{test}}, y) \leq \hat{q}\}$
 I153: $\mathbb{P}(Y_{\text{test}} \in C(X_{\text{test}})) \geq 1 - \alpha$
 I154: $h = (x, y)$
 I155: $f(h) = f((x, y)) = s(x, y)$
 I156: $s(X_1, Y_1), \dots, s(X_n, Y_n)$
 I157: $k = \lfloor (n+1)\alpha - 1 \rfloor$
 I158: α
 I159: $\frac{n-k}{n}$
 I160: $\frac{n-k}{n} = \frac{n - \lfloor (n+1)\alpha - 1 \rfloor}{n} = \frac{n + \lceil -((n+1)\alpha - 1) \rceil}{n} = \frac{\lceil n - (n+1)\alpha + 1 \rceil}{n} = \frac{\lceil (n+1)(1-\alpha) \rceil}{n}$
 I161: $g_{N,k}^* = \hat{q}$
 I162: $L_x(n-k, k+1) = \sum_{j=n-k}^n \binom{n}{j} x^j (1-x)^{n-j}$
 I163: $i = n - j$

$$\text{I164: } I_x(n-k, k+1) = \sum_{i=0}^k \binom{n}{n-i} x^{n-i} (1-x)^{n-(n-i)} = \sum_{i=0}^k \binom{n}{i} x^{n-i} (1-x)^i$$

$$\text{I165: } \mathbb{P}(Y_{\text{test}} \in C(X_{\text{test}}) | \{(X_i, Y_i)\}_{i=1}^n) \geq 1 - \epsilon$$

$$\text{I166: } \beta \geq I_{1-\epsilon}(n-k, k+1) = \sum_{i=0}^k \binom{n}{i} \epsilon^i (1-\epsilon)^{n-i}$$